

Geospatial Augmented Reality Tourist System

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ABSTRACT

Most existing intelligent AR tourist systems focus on recommending or presenting information related to points of interest (POIs) that do not consider user-generated datasets and the user's preference leading to user dissatisfaction. This paper aims to develop an intelligent AR system applying tourist behavior recognition and user-generated content using volunteer geospatial information (VGI). For POIs recommendation, user-generated content has been modeled with the VIKOR and optimized with the behavior recognition method. For POIs information presentation in HoloLens, sentiment analysis was utilized for VGI data generation as well as a deep learning method was performed for POIs object detection and visualization.

Keywords: Geospatial-augmented reality, volunteered geographic information, tourist behavior recognition, user-generated content, sentiment analysis, HoloLens.

Index Terms: Human-centered computing – Human-computer interaction (HCI) – Interaction paradigms – Mixed/augmented reality

1 INTRODUCTION

One of the significant challenges in the existing intelligent AR systems is the lack of user satisfaction in the field of services provided for tourist POIs recommendation and presentation because these systems are not considered the POIs data generated by previous visitors' experiences who already visited those places. While these systems pay less attention to their results presenting covering users' preferences who demanded tourist services. Various research has been performed in the field of tourist POIs recommendation and presentation. For instance, for POIs recommendation, intelligent autonomous agents learn their behavior and automatically provide recommendations based on the user's context, or ontology-based systems used knowledge-based structures to model the tourism domain to obtain user satisfaction [1]. Rokhsaritalemi *et al.* presented an intelligent system based on user preferences using a rule-based structure and multi-criteria decision-making (MCDM) [2]. Although VGI and user-generated content have been presented in some research [3], they have not modeled the POIs recommendation. Additionally, research has been done in the field of POIs information presentation, such as the research that has been done using XR technology in HoloLens for a more appropriate navigation information visualization using visited objects interacting with

gestures [4], while in the system, the user-generated content of previous users' experiences is ignored. Hence, the purpose of this paper is to provide intelligent modelling using VIKOR and behavior recognition methods to recommend POIs, and for POIs presentation, a sentiment analysis-based VGI for user-generated content and an artificial intelligence method for POIs object detection and visualization were employed.

2 OUR APPROACH

Our proposed system consists of two services (Figure 1); the first service was used to recommend suitable POIs for tourism. This service applied the geospatial, VGI, and behavior data to create criteria selection and combination with the VIKOR method to create a rating score for POIs recommendation. This adaptive-based model considered behavior recognition recommendations based on the mood, the daily status, and the environment of the user were utilized. To do so, R language and ArcGIS were employed to perform spatial analyses. The second service aims to propose a model to present geospatial and VGI information-related visiting objects in mobile and HoloLens interfaces using C# programming language. Our system has the following advantages: In the recommender model, the combination of geospatial data and VGI lead to more flexibility in the system based on the characteristics of the user. Therefore, in choosing a POI, the system will have both Geospatial criteria, such as distance to the user's location and VGI data such as service status. In addition to this, the user's behaviour leads to the removal of the visited POI during the user's daily trip. The display model acts as the explore service, which displays different information related to POI such as the scenery of the park, the preferred object and trails, and the reason for choosing based on each user's views.

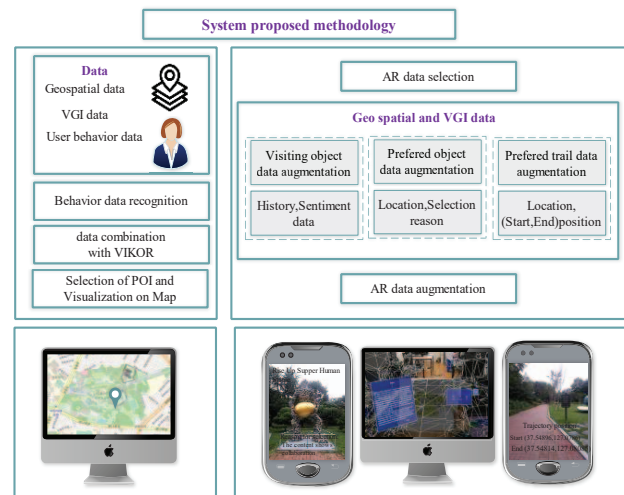


Figure 1: System methodology.

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2.1 Tourist POIs recommendation service

In this service, users' recommended POIs will be selected using the appropriate criteria by obtaining geospatial and user-generated data (VGI). Geospatial criteria include the distance to the user's location, the distance to the subway, and the price (entry fee) and VGI criteria contain the scenery, the feeling, and the service. After collecting data from users for one month, the three locations that have the highest number of VGI content suggested by each user were utilized for the modeling. Finally, all criteria related to POIs were weighted and combined with the VIKOR method to rank each POI in terms of tourist suitability. An analytic hierarchy process (AHP) approach based on the weight matrix (0.43, 0.03, 0.19, 0.07, 0.14, and 0.14) and the inconsistency index of 0.06 has been employed for fulfilling the weighting part inside the VIKOR method. To optimize the VIKOR method, during the behavior recognition step, the user's daily activities are retrieved in the system so that the user's visited places are eliminated from the VIKOR selection process, in this research, Seoul Forest was the final excluded POI after applying the behavior recognition method. Finally, the whole optimal VIKOR model chose Children's Grand Park as the first POI (Table 1 and Table 2).

Table 1. Tourist POI comparison; Where C1, C2, C3, C4, C5 and C6 are distance to User, Subway, Price, Scenery, Feeling, and services.

Name	C1	C2	C3	C4	C5	C6
Children's Grand Park	716.907	3352.203	9	8.5	7.5	8
Insadong	8037.563	259.761	9	8.5	9	7.5
LotteWorld	4877.295	6815.944	1	8	8	7

Table 2. Comparison of POI by using VIKOR analysis.

Item	Q ₁ V=0.5	Q ₂ V=1	Q Rank	S	S Rank	R	R Rank
1	0	0	1	0.014	1	0.014	1
2	0.759	0.519	2	0.43	2	0.43	2
3	0.776	1	3	0.814	3	0.244	3

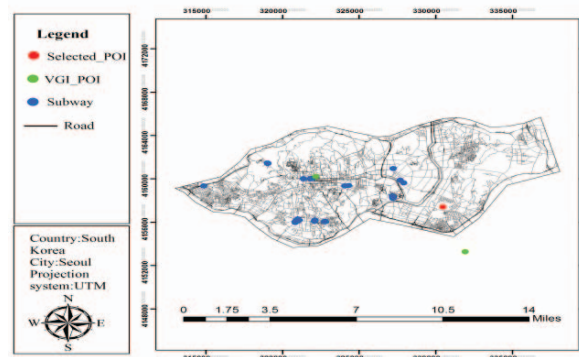


Figure 2: Tourist POI recommendation.

2.2 Tourist POI display component

The POIs presentation service visualizes VGI and geospatial data for tourists. This service is for each POI characteristic, a series of different information such as geospatial and VGI information produced by other users who had experience visiting that POI. For example, Figure 3 shows a 'park' POI in HoloLens presenting

some information such as the name, history, the user's comments, the number of positive sentiments about the VGI data related to the scenery of the park, and the average score of these users.



Figure 3: Tourist POI information display.

2.3 Conclusion and discussion

In this research, a model has been presented to recommend and Present tourism POIs in a way that leads to user satisfaction. The reason for using the behavior recognition method to optimize the VIKOR method is that without it, the final model could not handle complex scenarios such as the user's daily activities, location, and preferences. In addition, in future work, the user's surroundings and current situation parameters can be added to VIKOR modeling by adding biosensors information. The VIKOR-based MCDM method used in this research can model various and heterogeneous criteria compared to other Fuzzy rule-based methods that require increasing the number of rules by increasing the criteria. In the POIs information presentation service, we faced challenges to create a rating in the sentiment analysis, which can improve the results by designing a semantic structure. It was also concluded that deep learning models such as Yolo focus on processing speed and to achieve processing accuracy, attention should be paid to other deep learning methods such as Fast Region-based Convolutional. Network method (Fast R-CNN).

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